Shower Shape Comparison: Nominal vs Double-Interaction Mixed MC

1.5 mrad Crossing Angle (7.2% Pileup)

PPG12 Analysis Team (automated report)

April 9, 2026

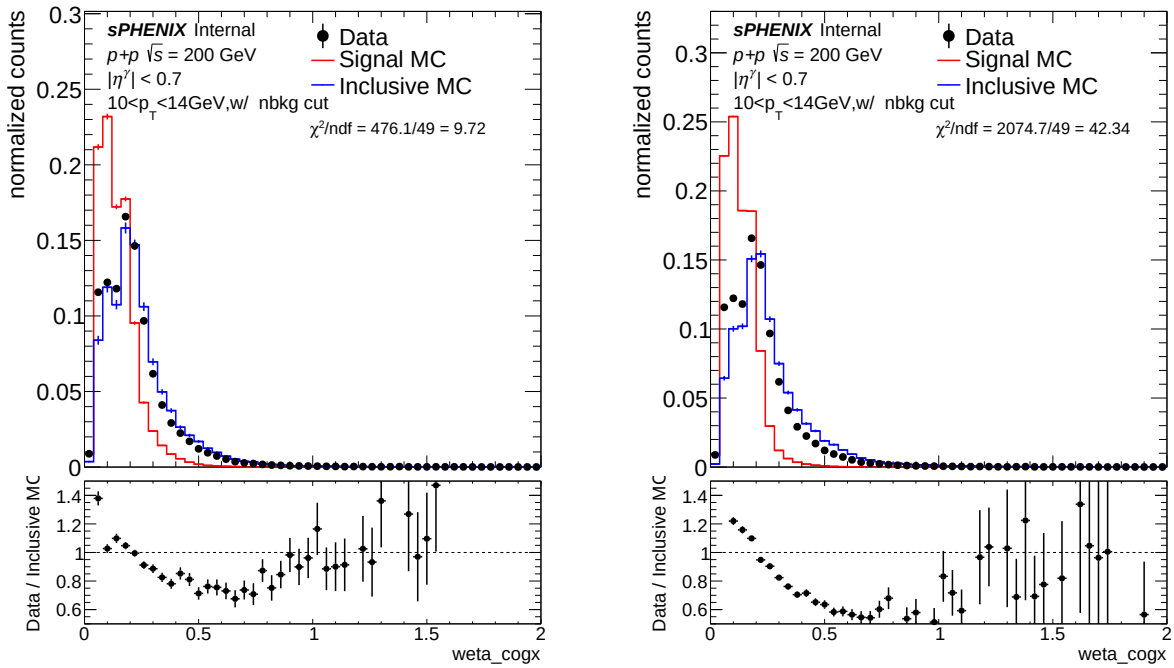
1 Introduction

This report extends the double-interaction shower shape comparison to the **1.5 mrad crossing-angle period** (runs 51274–54000, $\mathcal{L} = 16.9$ pb^{-1}), where the physics-dictated double-interaction fraction is **7.2%** — roughly 2.6× smaller than the 18.7% fraction at 0 mrad. The same two-pass blending pipeline is used: Pass 1 builds a combined vertex distribution; Pass 2 reruns the full shower shape analysis with consistent vertex reweighting and `mix_weight` applied to all histogram fills. The final "mixed MC" is the `hadd` of single (weight 0.928) and double (weight 0.072) output files.

The companion 0 mrad report demonstrated dramatic χ^2/ndf improvements (up to 22×) from the mixed MC. With only 7.2% pileup admixture at 1.5 mrad, the expected improvement is smaller, and the balance between correction and noise from the mixing procedure becomes more delicate.

2 Shower Shape Comparisons

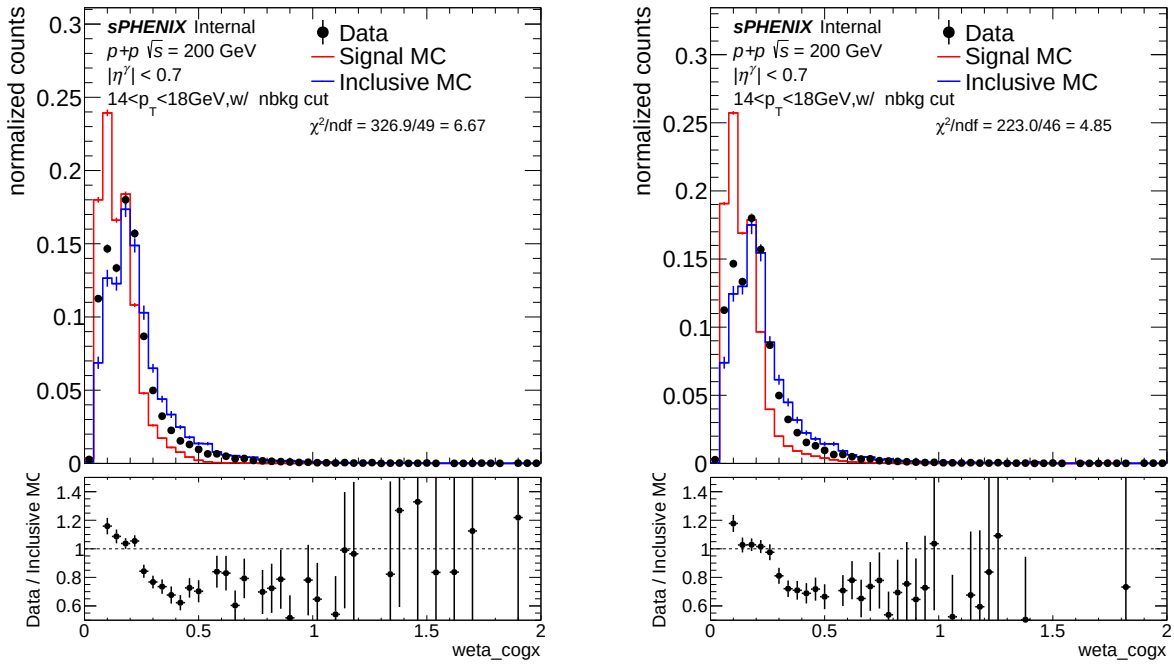
2.1 w_η^{CoG} (Eta Cluster Width)



(a) Nominal MC

(b) Mixed MC

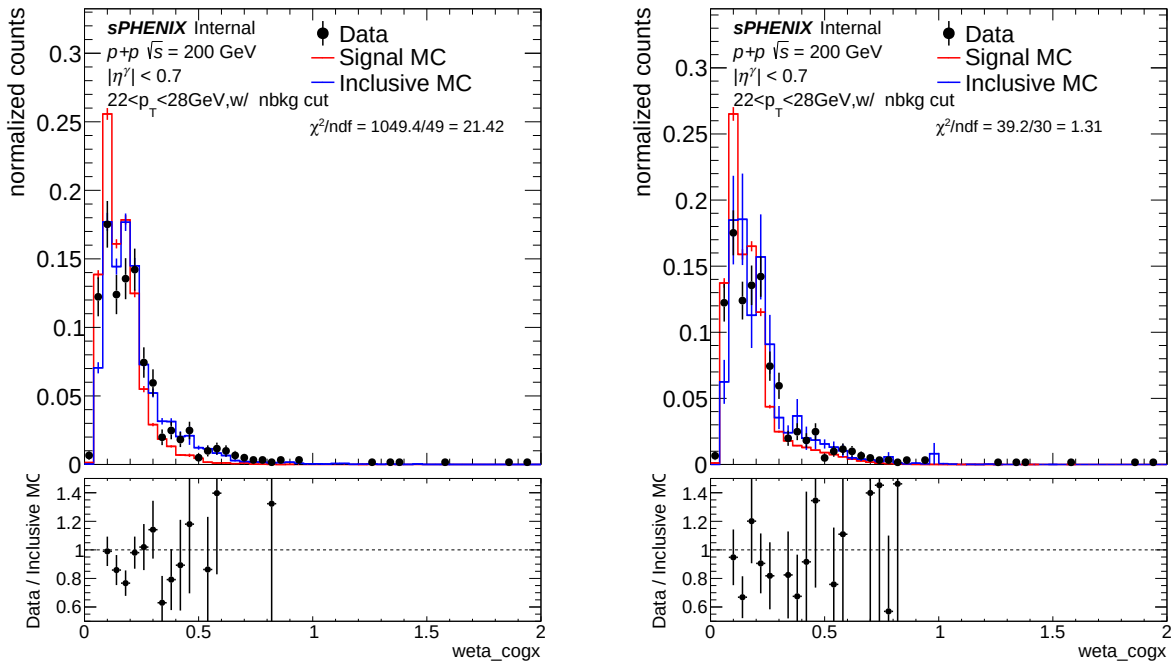
Figure 1: w_η^{CoG}, $10 < E_T < 14\,\mathrm{GeV}$ (pt0). Nominal $\chi^2/\mathrm{ndf} = 9.72$; mixed $\chi^2/\mathrm{ndf} = 42.34$. **The mixed MC degrades agreement by** $4.4\times$**.**



(a) Nominal MC

(b) Mixed MC

Figure 2: w_η^{CoG}, $14 < E_T < 18\,\mathrm{GeV}$ (pt1). Nominal $\chi^2/\mathrm{ndf} = 6.67$; mixed $\chi^2/\mathrm{ndf} = 4.85$. Modest $1.4\times$ improvement.



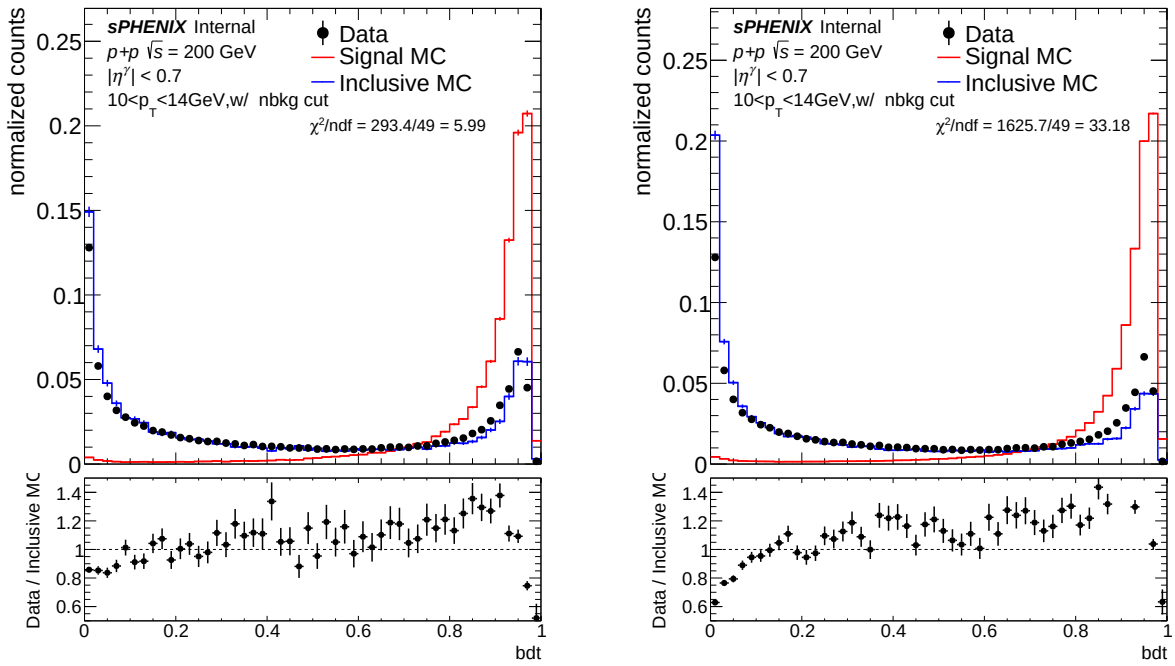
(a) Nominal MC

(b) Mixed MC

Figure 3: w_η^{CoG}, $22 < E_T < 28\,\mathrm{GeV}$ (pt3). Nominal $\chi^2/\mathrm{ndf} = 21.42$; mixed $\chi^2/\mathrm{ndf} = 1.31$. Strong 16.3× improvement.

The w_η^{CoG} behavior at 1.5 mrad shows a striking E_T-dependent pattern: the mixed MC *degrades* agreement at low E_T (4.4× worse at pt0) but provides a strong 16.3× improvement at high E_T (pt3). At pt1 the improvement is marginal (1.4×). This contrasts with the 0 mrad case where improvement was seen across all E_T bins.

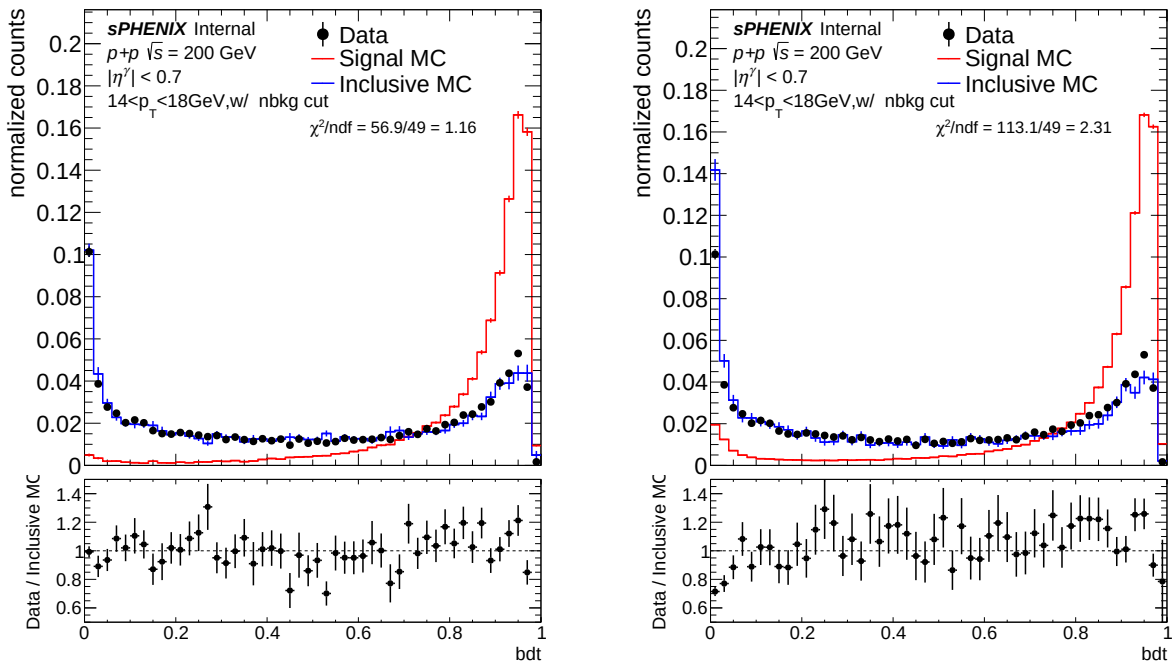
2.2 BDT Score



(a) Nominal MC

(b) Mixed MC

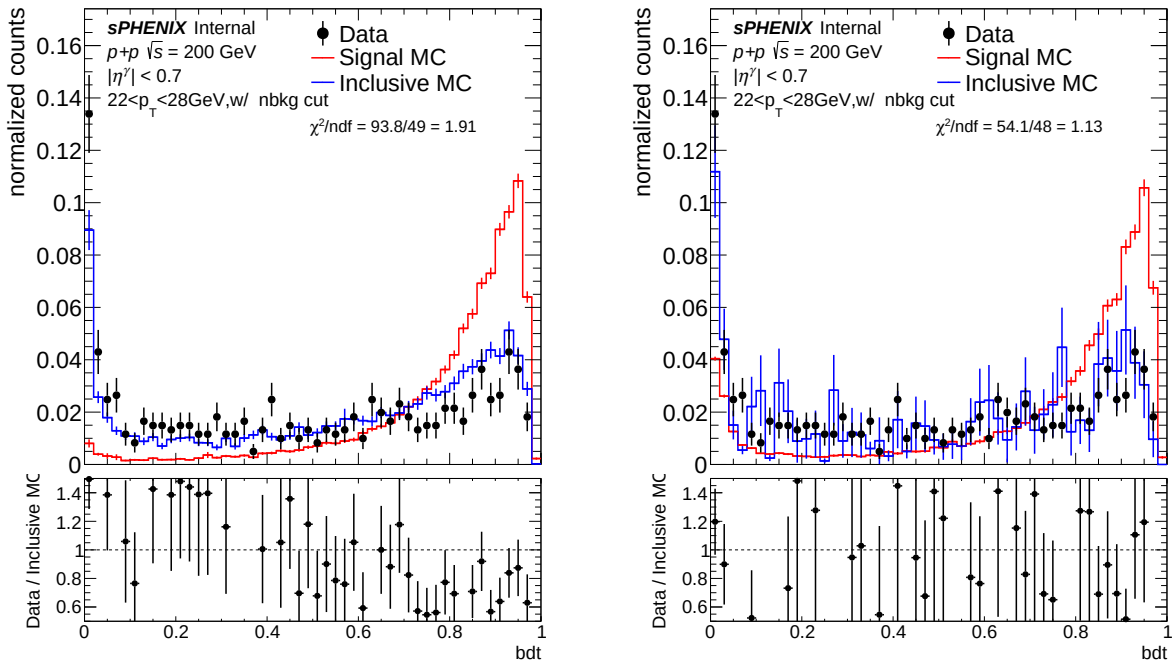
Figure 4: BDT score, $10 < E_T < 14\,\text{GeV}$ (pt0). Nominal $\chi^2/\text{ndf} = 5.99$; mixed $\chi^2/\text{ndf} = 33.18$. **Degradation by** $5.5\times$**.**



(a) Nominal MC

(b) Mixed MC

Figure 5: BDT score, $14 < E_T < 18\,\text{GeV}$ (pt1). Nominal $\chi^2/\text{ndf} = 1.16$; mixed $\chi^2/\text{ndf} = 2.31$. Degradation by $2.0\times$; nominal is already excellent.



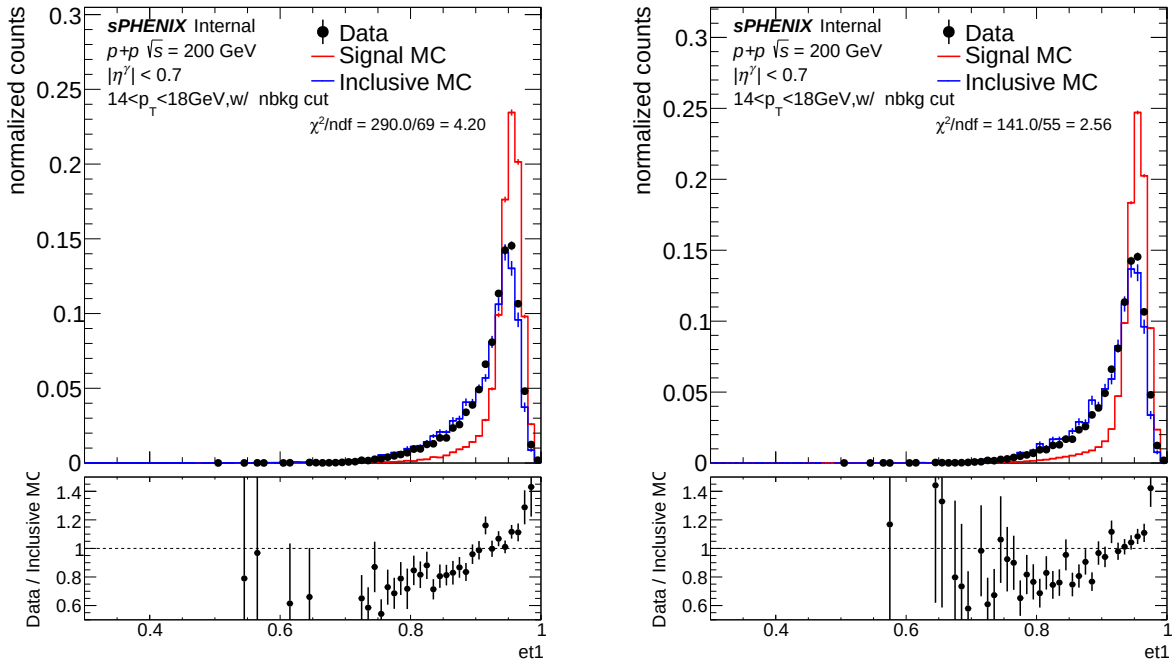
(a) Nominal MC

(b) Mixed MC

Figure 6: BDT score, $22 < E_T < 28\,\text{GeV}$ (pt3). Nominal $\chi^2/\text{ndf} = 1.91$; mixed $\chi^2/\text{ndf} = 1.13$. Modest $1.7\times$ improvement.

The BDT score at 1.5 mrad shows the same low-E_T degradation pattern: $5.5\times$ worse at pt0, $2.0\times$ worse at pt1 (where the nominal agreement is already excellent at $\chi^2/\text{ndf} = 1.16$), and a modest improvement at pt3 ($1.7\times$). The BDT amplifies the underlying shower shape distortions, making it particularly sensitive to the mixing procedure.

2.3 f_1 (Inner Ring Energy Fraction)



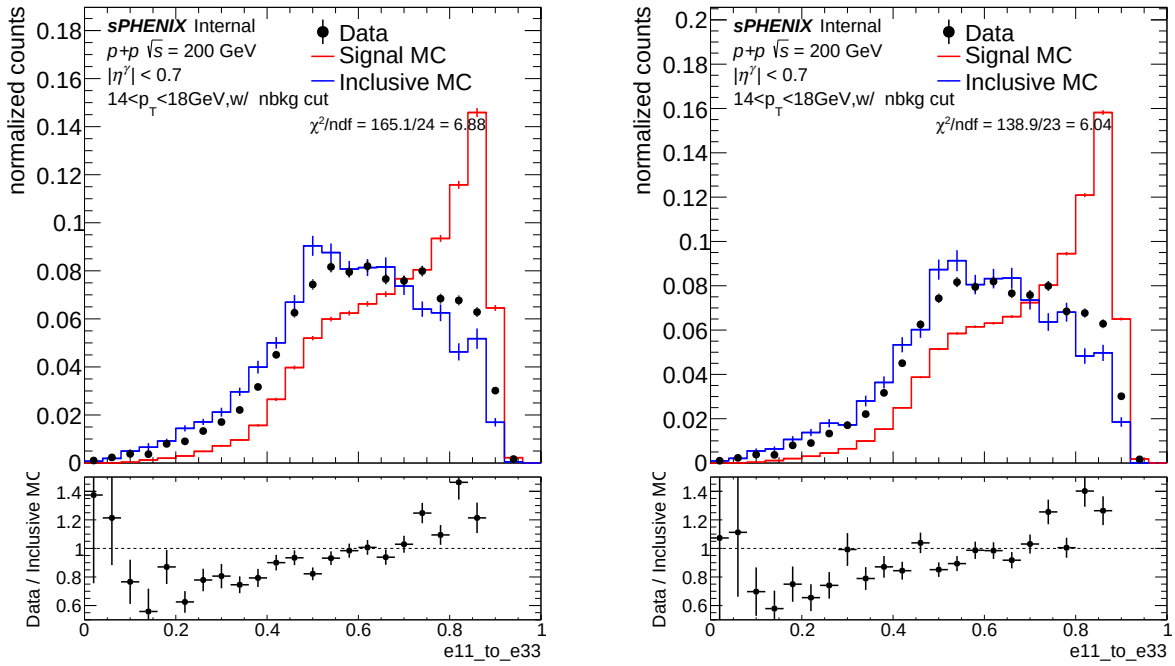
(a) Nominal MC

(b) Mixed MC

Figure 7: f_1, $14 < E_T < 18\,\text{GeV}$ (pt1). Nominal $\chi^2/\text{ndf} = 4.20$; mixed $\chi^2/\text{ndf} = 2.56$. $1.6\times$ improvement.

At pt1, f_1 shows a $1.6\times$ improvement (χ^2/ndf from 4.20 to 2.56), comparable to the improvement seen for w_η^{CoG} at the same E_T bin.

2.4 $E_{1\times1}/E_{3\times3}$ (Seed/Core Ratio)



(a) Nominal MC

(b) Mixed MC

Figure 8: $E_{1\times1}/E_{3\times3}$, $14 < E_T < 18\,\text{GeV}$ (pt1). Nominal $\chi^2/\text{ndf} = 6.88$; mixed $\chi^2/\text{ndf} = 6.04$. Marginal $1.1\times$ improvement.

$E_{1\times1}/E_{3\times3}$ shows only a marginal improvement (χ^2/ndf from 6.88 to 6.04 at pt1), consistent with the 0 mrad finding that this ratio is less sensitive to double-interaction effects.

3 χ^2/ndf Summary

Table 1 compiles the Data–Inclusive MC χ^2/ndf values for both the 1.5 mrad and 0 mrad cases. Entries where the mixed MC degrades agreement are highlighted in red.

Table 1: Data–Inclusive MC χ^2/ndf for nominal (single-only) and mixed (double-blended) MC at cut1. 1.5 mrad uses 7.2% double fraction; 0 mrad uses 18.7%. Lower is better.

		1.5 mrad (7.2%)		0 mrad (18.7%)	
Variable	E_T bin [GeV]	Nominal	Mixed	Nominal	Mixed
w_η^{CoG}	10–14 (pt0)	9.72	42.34	47.37	29.08
w_η^{CoG}	14–18 (pt1)	6.67	4.85	17.63	2.99
w_η^{CoG}	22–28 (pt3)	21.42	1.31	16.88	0.75
BDT score	10–14 (pt0)	5.99	33.18	—	—
BDT score	14–18 (pt1)	1.16	2.31	12.18	1.63
BDT score	22–28 (pt3)	1.91	1.13	—	—
f_1	14–18 (pt1)	4.20	2.56	9.33	1.93
$E_{1\times1}/E_{3\times3}$	14–18 (pt1)	6.88	6.04	4.51	4.14

4 Discussion

The 1.5 mrad results reveal a qualitatively different picture from the 0 mrad case. The key observation is an **E_T-dependent crossover**: at low E_T the mixed MC makes agreement worse, while at high E_T it provides clear improvement.

4.1 Low-E_T degradation

At pt0 (10–14 GeV), the mixed MC degrades χ^2/ndf by 4.4× for w_η^{CoG} and 5.5× for the BDT score. Several factors contribute:

1. **Small pileup fraction amplifies noise.** At 7.2% mixing, the double-interaction MC contributes only a small correction. Any mismatch between the double MC shower shapes and the actual pileup characteristics of the 1.5 mrad data period (different luminosity profile, bunch structure) gets amplified relative to the small physical signal.

2. **Shared MC samples.** The same `photon10_double` and `jet12_double` samples are used for both crossing angles — only the mixing weight differs. The double-interaction MC was produced with beam conditions that may better match the 0 mrad period. Producing crossing-angle-specific double-interaction MC could resolve this.

3. **Vertex distribution mismatch.** The two-pass vertex reweighting uses the combined (blended) vertex distribution. At 7.2% double fraction, this combined distribution is very close to the single-interaction distribution, so any systematic offset in the double MC vertex shape has an outsized effect per double-interaction event.

4. **Low-E_T clusters are most fragile.** Clusters near the 8–14 GeV threshold are most sensitive to vertex shifts and energy redistribution, making them a worst-case test of the mixing procedure.

4.2 High-E_T improvement

At pt3 (22–28 GeV), the mixed MC improves w_η^{CoG} by 16.3× (21.42 → 1.31) and BDT by 1.7× (1.91 → 1.13). This indicates that double-interaction effects are still physically relevant at 1.5 mrad for high-E_T clusters, and the MC correction captures them well. High-E_T clusters are more robust to the mixing procedure because they are less affected by threshold effects and energy redistribution.

4.3 Comparison with 0 mrad

At 0 mrad (18.7% pileup), the mixed MC improved agreement across all E_T bins, with improvement factors growing from 1.6× at pt0 to 22.5× at pt3 for w_η^{CoG}. The 1.5 mrad case shows that the crossover from "correction helps" to "correction hurts" occurs somewhere between 7.2% and 18.7% pileup fraction at low E_T.

Notably, the nominal 1.5 mrad χ^2/ndf values are already much better than the 0 mrad nominals (e.g., w_η^{CoG} pt0: 9.72 vs. 47.37; BDT pt1: 1.16 vs. 12.18), confirming that lower pileup means less need for correction.

5 Conclusions and Recommendation

1. The 7.2% double-interaction blending at 1.5 mrad **improves** data–MC agreement at high E_T ($\geq$ 18 GeV), with up to 16× improvement in w_η^{CoG} at pt3.

2. At low E_T ($< 14\,\text{GeV}$), the blending **degrades** agreement by up to $5.5\times$, likely due to the small pileup fraction amplifying MC mismodeling and the sensitivity of low-E_T clusters to the mixing procedure.

3. For the cross-section measurement, the 1.5 mrad data should **not** use the mixed MC as baseline at low E_T. The recommended approach:

   - $E_T \geq 18\,\text{GeV}$: use mixed MC (7.2% double-blended) as baseline
   - $E_T < 18\,\text{GeV}$: use nominal (single-only) MC as baseline; assign the difference between nominal and mixed as a systematic uncertainty

4. Alternatively, the full analysis could use the nominal MC for the 1.5 mrad period (where pileup is inherently small) and reserve the double-interaction correction for the 0 mrad period only. The nominal agreement at 1.5 mrad is already acceptable ($\chi^2/\text{ndf} \lesssim 10$ for most variables).